

4(a)	extension / <u>original</u> length	B1			
4(b)(i)	Young modulus = stress / strain	C1			
	$= (18 / 4.5 \times 10^{-7}) / (1.4 \times 10^{-3} / 4.0)$	C1			
	$= (4.0 \times 10^7) / (3.5 \times 10^{-4})$ $= 1.1 \times 10^{11} \text{ Pa}$	A1			
4(b)(ii)	straight line through the origin	B1			
	ending at the point (3.5, 4.0)	B1			
4(c)			B2		
		greater in second wire		less in second wire	the same in both wires
	stress			✓	
	strain		✓		